Mathematical model

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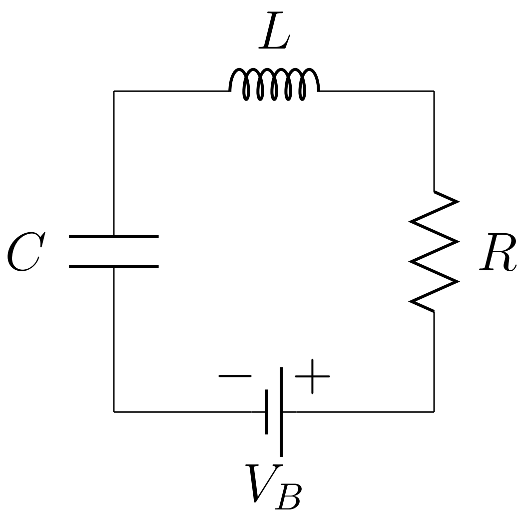
For a base clock for a computer, we want to build an *LC-oscillator* that generates 100 MHz oscillations.



The components that are given are:

- a 5V battery: $V_B = 5$ V,
- an inductor: $L = 0.004\,\mu$H,
- the resistance: $R = 0.1\,\Omega$.

Our problem is

> *Find the capacitance C of the capacitor ($[C] =$ henry) such that the current in the electrical circuit oscillates with a frequency of 100 MHz.*

The following physical laws relate the current $I(t)$ running in the circuit (measured in ampere), the charge on (one side of) the capacitor $Q(t)$ (in coulomb) and the voltage drops over the different elements (in volts).

The current, $I(t)$, is related to the charge on the capacitor, $Q(t)$, according to

$$I(t) = \frac{dQ}{dt}.$$

The voltage drop over the capacitor, $V(t)$, is related to the charge on the capacitor by:

$$V(t) = \frac{Q(t)}{C}.$$

The voltage drop over the inductor, $V_L(t)$, is related to the change in the current:

$$V_L(t) = L\frac{dI}{dt}.$$

And an expression for voltage drop over the resistor, $V_R(t)$ is given by: $V_R(t) = RI(t)$.

Kirchhoff's voltage law states that the voltage drop around a loop in a circuit equals zero:

$$V + V_L + V_R - V_B = 0.$$

There are different options for a set of dependent variables to calculate in. We choose to calculate in current $I(t)$ and the voltage drop over the capacitor $V(t)$.

Exercise A

1/1 point (graded)

Eliminate the charge from the first two physical laws and derive a relation between V and I :

- ( ) $\frac{dV}{dt} = I(t)$
- ( ) $C\frac{dI}{dt} = V(t)$
- (•) $C\frac{dV}{dt} = I(t)$ ✔

Submit You have used 1 of 1 attempt

Answers are displayed within the problem

Exercise B

1/1 point (graded)

Use the last three physical laws to derive the second relation between V and I:

- (selected, correct) $L\frac{dI}{dt} = V_B - R\,I(t) - V(t)$ ✓
- $\frac{L}{C}\frac{dI}{dt} = CV_B - R\,I(t) - V(t)$
- $L\frac{dI}{dt} = \frac{V_B}{C} + R\,I(t) + V(t)$

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Exercise C

1/1 point (graded)

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PROBLEM

Combine the two differential equations from exercises A and B into one system of equations of the form shown below.

$$\frac{d\vec{X}}{dt} = AX + \vec{b},$$

with $\vec{X} = \begin{bmatrix} I \end{bmatrix}$,

$$A = \begin{bmatrix} 0 & 1/C \\ -1/L & -R/L \end{bmatrix},$$

and $\vec{b} = \begin{bmatrix} V_B/L \\ \; \end{bmatrix}$.

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FEEDBACK

✔ Correctly placed 6 items.

✔ Good work! You have correctly formulated the vector differential equation.

✔ Your highest score is 1.0

Initial conditions

We will assume that before $t = 0$, there is no current, and the capacitor is not charged. Only at $t = 0$, the battery is connected. The initial conditions are then

$$V(0) = 0, \quad I(0) = 0.$$

Eager to start simulating? On the next page, first a scaling, and then you can start up Euler's method!

English ▼

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